ANOMALY DETECTION IN NETWORK TRAFFIC USING UNSUPERVISED AND SUPERVISED LEARNING ON CICIDS2017

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Abstract: This paper investigates the effectiveness of various machine learning paradigms in detecting network intrusions using the benchmark CICIDS2017 dataset. We implement a comprehensive pipeline including data cleaning, dimensionality reduction via Incremental PCA, and a comparative analysis between unsupervised methods (One-Class SVM, Local Outlier Factor) and supervised learning (Multi-layer Perceptron). Our experimental results demonstrate that while supervised models achieve superior accuracy on known threats, unsupervised boundary-based methods like OCSVM exhibit greater robustness when generalizing to unknown attack patterns.

Keywords: *INTRUSION DETECTION, CICIDS2017, MACHINE LEARNING, ANOMALY DETECTION, PCA.*

I. INTRODUCTION

The rapid growth of the internet has changed how we live and work, but it has also led to a massive increase in complex cyberattacks. As these digital threats get smarter, we need better security systems to protect our private information and keep networks running safely. The main goal of this project is to explore how machine learning can help by telling the difference between normal network traffic and malicious activity. By comparing supervised and unsupervised learning models, this study looks for the best way to defend against both well-known attacks and new, "zero-day" threats that haven't been seen before.

II.RELATED WORK

My research builds on several studies that use the CICIDS2017 dataset to improve network security. A key study by Xu and Liu [2] compared models like MLP and OCSVM, finding that while deep learning is perfect for known attacks, it fails to recognize new threats, which they call the "closed-world" problem. This highlights why testing for "zero-day" robustness is so important.

Additionally, Prasad et al. [3] emphasize that as cyberattacks become more complex, traditional signature-matched defenses are no longer enough, making anomaly-based machine learning a necessity for modern protection. Finally, Oyelakin et al. [4] analyzed the CICIDS2017 dataset itself, noting that its massive size and class imbalance make it a challenging benchmark. Their findings justify the use of specialized tools like Incremental PCA to handle such a large volume of data efficiently.

III. METHODOLOGY

A. Network Intrusion Detection

An Intrusion Detection System (IDS) is a tool that monitors network traffic to find suspicious activity or security breaches. Traditional systems usually use Signature-based detection (S-IDS), which works by comparing network activity against a database of "signatures" or known attack patterns. While S-IDS is very accurate at catching threats it already knows, it cannot identify "Zero-Day" attacks because their patterns haven't been recorded yet.

To fix this, this project focuses on Anomaly-based detection (A-IDS). Instead of looking for specific fingerprints, A-IDS works by creating a "baseline" of what normal network behavior looks like. If the system sees a major change from this baseline, it flags it as a potential attack. As noted in recent research, this ability to spot unusual behavior is why anomaly detection is so effective at finding new threats that haven't been seen before.[1]

B. The CICIDS2017 Dataset

For the experimental evaluation, this study uses the CICIDS2017 dataset, which is a widely recognized benchmark for testing intrusion detection systems. Unlike older datasets, CICIDS2017 is much more realistic because it consists of real network traffic captured over five days of activity.

The dataset is technically complex, containing over 2830743 records and 79 statistical features, such as flow duration and packet length. These records are categorized into 15 different labels, ranging from "Benign" (normal traffic) to various malicious attacks. However, working with this data is difficult due to its massive scale and a severe class imbalance, where normal traffic far outweighs the number of attack samples. This imbalance is why specialized preprocessing and sampling techniques were necessary for this project.[3]

C. DATA PREPARATION

Before training the models, I had to transform the raw data into a format that machine learning algorithms can understand. This started by merging eight different files into one large table and cleaning it by fixing headers and removing 308381 duplicate rows.

Because the dataset was so large, it used a lot of computer memory (1539.53 MB). To fix this, I used a "downcasting" technique to change the data types from 64-bit to 32-bit, which cut the memory usage down to 808.25

MB a 47.50% improvement in efficiency. I also handled "infinite" values by replacing them with the median, because the median isn't easily swayed by the extreme peaks usually seen in network traffic.[2]

To make the system even faster, I removed features that never changed and used a method called Incremental Principal Component Analysis (IPCA). This allowed me to cut the number of features in half while still keeping 99.30% of the original information. Using IPCA is particularly effective for large datasets like CICIDS2017 because it processes data in small batches rather than all at once.[1]

Metric	Raw Dataset	Processed Dataset
Total Rows	2830743	2522362
Number of Features	79	35
Memory Usage (MB)	1539.53	808.25
Duplicate Records	308381	0
Null/Inf Values	3128	0

Figure 1 Impact of Data Cleaning and Memory Optimization

D. EXPERIMENTAL DESIGN AND DATA SPLITTING STRATEGY

The core of this research lies in its specific data partitioning strategy, designed to evaluate model robustness against both anticipated and unanticipated network threats. Unlike standard random splits, this study implements a controlled isolation of specific attack vectors namely DoS slowloris, DoS Slowhttptest, and Bot which are designated as "unknown attacks". By excluding these labels from the training phase of the supervised models, the experiment creates a rigorous "zero-day" simulation to test whether deep learning can generalize beyond its explicit training distribution.

The data splitting logic is divided into three primary functional blocks:

Supervised Training Set Construction: A logical mask is applied to the master dataframe to filter out the three "unknown" attack types. The remaining data, consisting of benign traffic and eleven known attack categories, is split using an 80/20 ratio. Stratified sampling is employed to ensure that the class distribution of benign versus known malicious traffic remains consistent between the training (X_train_supervised) and verification sets.[1]

Unsupervised Baseline Generation: To train the anomaly detection models (OCSVM and LOF), a dedicated training set (X_train_unsupervised) is created by sampling 200,000 records exclusively from the "BENIGN" class. This ensures the models learn a pure representation of "normal" network behavior without any exposure to malicious signatures.[1]

Zero-Day Evaluation Framework: A specialized test set (X_test_unknown) is constructed to evaluate model performance on novel threats. This set combines the previously isolated unknown attacks with an equal number of benign samples. By balancing the classes in this subset, the study provides an unbiased measure of the models' ability to flag truly novel anomalies in a controlled, binary-classification environment.[1]

E. CLASSIFIERS

In this phase, I tested three different types of machine learning models to see how well they identify known attacks and how they handle new, unknown threats. By using a Multi-Layer Perceptron (MLP), a One-Class Support Vector Machine (OCSVM), and a Local Outlier Factor (LOF), I can compare supervised deep learning against unsupervised methods.

Multi-Layer Perceptron (MLP): The MLP is a supervised learning model, which means it learns by looking at labeled examples of both "Benign" and "Attack" traffic. The structure I built has an input layer, two hidden layers (with 100 and 50 neurons), and an output layer that gives a final "yes or no" for an attack. It uses the "ReLU" activation function to understand complex patterns and the Adam optimizer to learn efficiently. While the MLP is great at catching attacks it has seen before, it often fails to recognize new threats because it only knows what it was explicitly taught this is known as the "closed-world" problem.[1]

One-Class Support Vector Machine (OCSVM): To catch "Zero-Day" attacks, I used the OCSVM. Unlike the MLP, this model is unsupervised and only trains on normal (benign) data. It works by drawing a boundary, or a "sphere of normalcy," around the normal data points. Anything that falls outside this boundary is flagged as an attack. I used a specific setting (RBF kernel with $\nu = 0.05$) to make sure the boundary is tight enough to catch hackers but flexible enough not to cause too many false alarms.

Local Outlier Factor (LOF): The LOF is another unsupervised method, but it focuses on density rather than boundaries. It looks at a data point and compares it to its 80 nearest neighbors. If a point is in a much thinner, less dense area than its neighbors, the model flags it as an outlier (an attack). This is very useful for finding strange behaviors that don't look like any known attack but are clearly different from normal traffic.

F. PERFORMANCE METRICS AND EVALUATION

To quantitatively assess the effectiveness of the selected classifiers, the study employs a comprehensive suite of evaluation metrics derived from the standard confusion matrix(Figure 2). The matrix records four fundamental outcomes: True Positives (TP) and True Negatives (TN), representing correct predictions of malicious and benign traffic respectively, and False Positives (FP) and False Negatives (FN), which indicate misclassifications. These values are used to calculate Accuracy(1), representing the overall ratio of correct predictions to total samples, and the F1-Score(4), which serves as the harmonic mean of Precision(2) and Recall(3) to provide a balanced measure of performance on the imbalanced CICIDS2017 dataset.

In the context of network security, specific emphasis is placed on Precision and Recall. Precision measures the reliability of the system by determining the proportion of flagged alerts that are actual attacks, thereby quantifying the "false alarm" rate. Recall, or sensitivity, is critical for measuring the detection coverage, as it defines the model's ability to identify all malicious instances within the traffic stream. Furthermore, the Receiver Operating Characteristic Area Under the Curve (ROC-AUC) is

calculated for each model. This metric is particularly valued in this study as it evaluates the model's discriminative capability across various decision thresholds, providing a robust indicator of how well a classifier separates benign traffic from anomalous activities regardless of the specific class distribution.

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Figure 2 Confusion Matrix

$$A = \frac{True\ positive + \text{True Negative}}{TP + FN + FP + TN} \quad (1)$$

$$P = \frac{True\ positive}{True\ positive + False\ positive} \quad (2)$$

$$R = \frac{True\ positive}{True\ positive + False\ negative} \quad (3)$$

$$F_1 = 2\frac{P \times R}{P + R} \quad (4)$$

IV. RESULTS

The evaluation of the implemented Intrusion Detection Systems (IDS) focuses on two critical dimensions: the ability to recognize established attack patterns and the capacity to generalize to novel, "zero-day" threats.

A. Analysis of Known Attack Detection

In the first evaluation scenario involving the Overall Test Set, the supervised Multi-Layer Perceptron (MLP) demonstrated superior discriminative accuracy. The model achieved a ROC-AUC score of 0.9936, correctly identifying 82252 malicious samples while maintaining a precision of 0.99. This performance is attributed to the model's exposure to 11 distinct attack labels during the training phase, allowing it to map high-dimensional features directly to specific malicious profiles. In comparison, the unsupervised models yielded lower precision on familiar traffic, with the Local Outlier Factor (LOF) achieving a ROC-AUC of 0.887 and the One-Class SVM (OCSVM) reaching 0.757. These results confirm that while unsupervised methods provide a wider safety net, they lack the fine-grained precision of supervised learning when attack signatures are previously known.

B. Evaluation of Zero-Day Robustness

The second scenario utilized the X_test_unknown dataset, specifically designed to simulate a real-world environment where Bot, DoS slowloris, and DoS Slowhttptest attacks appear as novel anomalies. Under these conditions, the performance of the MLP collapsed significantly, with its recall dropping to 0.15. This failure highlights the "closed-world" limitation of supervised models, which misclassify unknown threats as benign traffic because they lack a corresponding training label.

Conversely, the unsupervised models proved to be more effective for zero-day detection. The OCSVM achieved the most balanced performance in this scenario, maintaining a recall of 0.62 and a high precision of 0.92. By learning the "sphere of normalcy" from 200000 benign records, the OCSVM successfully flagged novel attacks as outliers without ever having seen their specific signatures.

Similarly, the LOF model provided a recall of 0.41, outperforming the supervised baseline and demonstrating that density-based isolation is a viable secondary layer for anomaly detection.

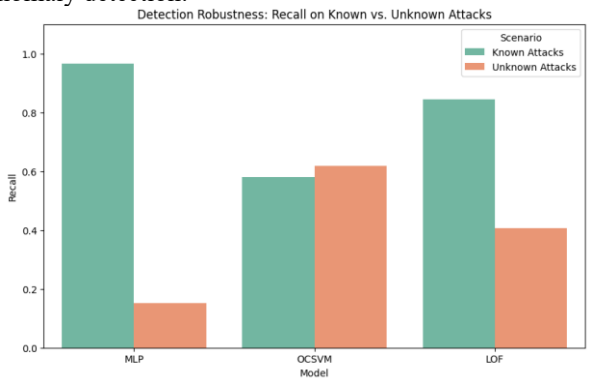

Figure 3 Bar Chart Comparing Recall and F1-Score for All Models

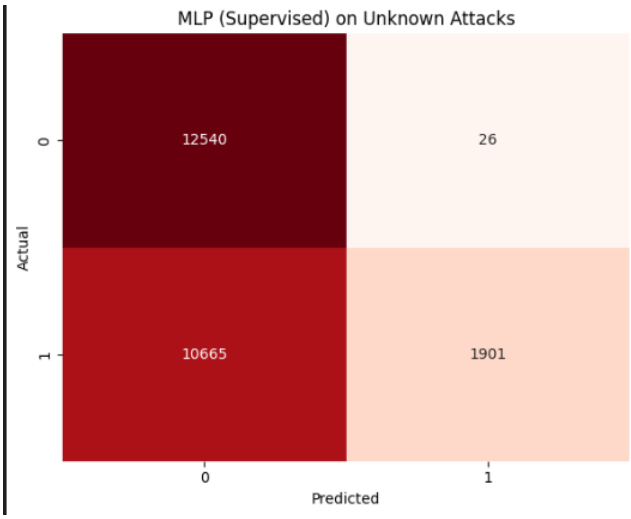

Figure 4 Confusion matrix MLP on Unknown Attacks

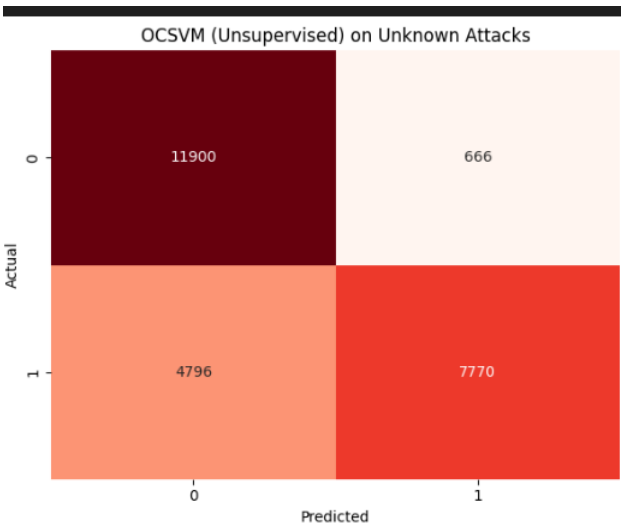

Figure 5 Confusion matrix on OCSVM on Unknown Attacks

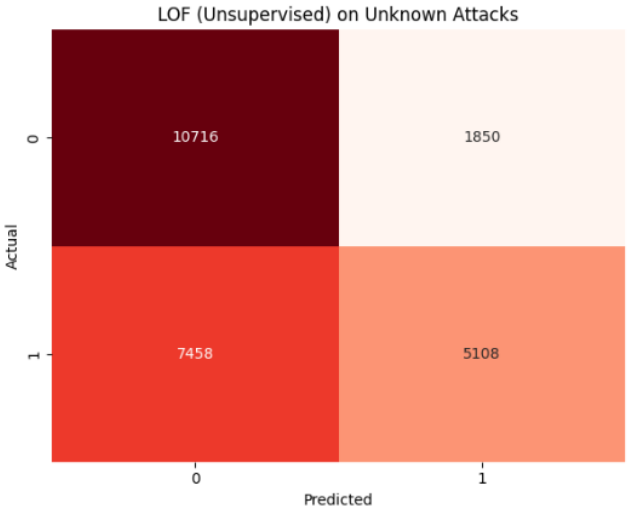

Figure 6 Confusion matrix LOF on Unknown Attacks

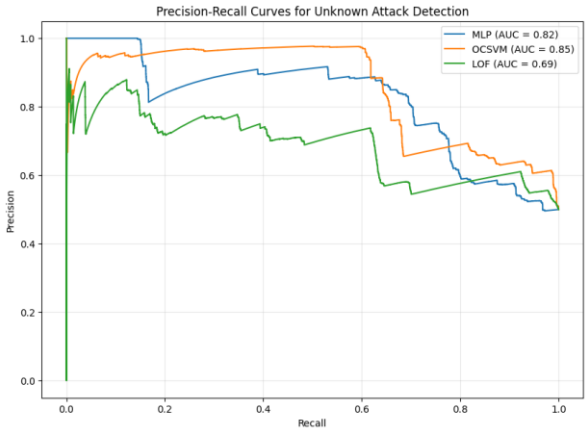

Figure 7 Precision-Recall Curves for Unknown Attack Detection

C. Summary of Findings

The results underscore a fundamental trade-off in network security: supervised models offer high reliability for repetitive threats, but unsupervised boundaries are necessary to mitigate the risks of evolving cyber-attacks. The OCSVM's ability to maintain a recall of 0.62 on completely novel patterns, compared to the MLP's 0.15, justifies its role as a critical component in robust, modern IDS architectures.

Model	Scenario	Accuracy	Precision	Recall	F1
MLP	Known Attacks	0.99	0.99	0.97	0.98
	Unknown Attacks	0.57	0.99	0.15	0.26
OC-SVM	Known Attacks	0.89	0.68	0.58	0.63
	Unknown Attacks	0.77	0.92	0.62	0.74
LOF	Known Attacks	0.85	0.54	0.84	0.65
	Unknown Attacks	0.62	0.73	0.41	0.52

Figure 8 Summary table

V. CONCLUSIONS

This study evaluated the performance of supervised and unsupervised machine learning models using the CICIDS2017 dataset, focusing specifically on the challenge of detecting "zero-day" network threats. The results highlight a major difference in how these models handle unknown traffic. While the Multi-Layer Perceptron (MLP) was nearly perfect at catching known attacks achieving 0.99 accuracy and a 0.9936 ROC-AUC its performance dropped to a recall of only 0.15 when faced with new, unseen attacks. This proves that while supervised models are excellent for stopping familiar threats, they struggle to generalize when a hacker uses a brand-new method that wasn't in the training data.

In contrast, the One-Class Support Vector Machine (OCSVM) proved to be much more reliable for finding these novel attacks. By learning a "sphere of normalcy" from 200000 benign records, it successfully flagged unknown threats with a recall of 0.62 and a precision of 0.92. Additionally, the preprocessing steps were critical for the project's success; using memory downcasting and Incremental PCA reduced the memory footprint by 47.50% while keeping 99.30% of the original data variance.

Ultimately, this research suggests that a truly robust Intrusion Detection System should not rely on supervised learning alone. Instead, it should include unsupervised components to act as a safety net for novel vulnerabilities. Future work should focus on building hybrid systems that combine the high precision of supervised deep learning with the flexible detection capabilities of unsupervised models.

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